MATH 231 – Linear Algebra I

The Vector Unit, Part 5 ▪ Orthogonality, Independence, and Subspaces

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About this document. This is Part 5 of the vector unit. Section numbering runs continuously through all parts, so a reference such as §18.5 points into Part 4 and means exactly what it says:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms
4	§16–§20	the standard basis, span, coordinates, $\mathbb{R}^3$
5	§21–	orthogonality, independence, subspaces

Assumed from Parts 1–4. The dot product and the orthogonality test $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (§11.7); the identity $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$ (§11.6); projections (§12.4); span, basis, coordinates and dimension (§16–§17); linear independence (§18.4) and the fact that an independent pair spans $\mathbb{R}^2$ (§18.5); subspaces (§19).

What you should be able to do afterwards. State and use linearity of the dot product; prove that mutually orthogonal nonzero vectors are linearly independent, both geometrically and algebraically; state the lemma that such vectors span; explain why the converse fails; compute coordinates against an orthogonal basis by projection alone; describe the subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$ as lines and planes through the origin; and prove the Pythagorean identity and use it to account exactly for a projection.

21. Orthogonality and independence

Part 4 asked which pairs of vectors form a basis and answered: the independent ones. But independence is a slightly awkward condition to check — you set up a combination equal to $\mathbf{0}$ and solve a system. This section shows there is a sufficient condition that is far easier to test, one we have had since §11.7.

21.1. What we will use: the dot product is bilinear

Until now most dot-product computations have been done by expanding into coordinates; §12.6 verified an orthogonality claim that way on purpose. The argument in §21.2 is different, because it is the one that keeps working in dimensions we cannot draw, and instead of coordinates it uses the algebraic laws collected in §11.8. The piece it needs is bilinearity, in this combined form:

$$\big\langle \mathbf{u},\ \alpha\mathbf{v} + \beta\mathbf{w} \big\rangle = \alpha\langle \mathbf{u}, \mathbf{v} \rangle + \beta\langle \mathbf{u}, \mathbf{w} \rangle,$$

which is **D2** and **D3** applied together, and which by symmetry (**D1**) holds in the first argument as well. Recall also **D4**, $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$.

Worth noting once: none of the verifications in §11.8 mentioned how many coordinates a vector has, so every one of **D1**–**D4** holds verbatim in $\mathbb{R}^3$ and in $\mathbb{R}^n$. That is what makes the argument below dimension-independent, and it is why we reach for the laws rather than the components.

21.2. Orthogonal vectors are independent

Claim. If $\mathbf{u}$ and $\mathbf{v}$ are nonzero and orthogonal, then they are linearly independent.

The intuition

Take the geometric route first, because it is a single sentence. Two nonzero vectors are dependent exactly when they are parallel (§18.4), and parallel means the angle between them is $\theta = 0$ or $\theta = \pi$. Orthogonal means $\theta = \pi/2$. A pair cannot have two different angles, so an orthogonal pair cannot be a parallel pair, and therefore cannot be dependent.

Put less formally: dependence means one vector is a stretched copy of the other, so it lies entirely *along* it. Orthogonal vectors lie entirely *across* each other — neither has any component in the other's direction at all, which is what a zero dot product says. Stretching something that points due east will never produce something pointing due north.

The proof that survives into higher dimensions

The geometric argument leans on θ, which is a picture. Here is the same fact without one. Suppose

$$\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}.$$

Take the dot product of both sides with $\mathbf{u}$. The right side gives $\langle \mathbf{u}, \mathbf{0}\rangle = 0$, and the left side expands by §21.1:

$$\big\langle \mathbf{u},\ \alpha\mathbf{u} + \beta\mathbf{v}\big\rangle = \alpha\langle \mathbf{u}, \mathbf{u}\rangle + \beta\langle \mathbf{u}, \mathbf{v}\rangle = \alpha\|\mathbf{u}\|^2 + \beta \cdot 0 = \alpha\|\mathbf{u}\|^2,$$

using $\langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$ from §11.6 and orthogonality to kill the cross term. So $\alpha\|\mathbf{u}\|^2 = 0$; and since $\mathbf{u} \neq \mathbf{0}$ we have $\|\mathbf{u}\|^2 > 0$, forcing $\alpha = 0$. Dotting instead with $\mathbf{v}$ gives $\beta\|\mathbf{v}\|^2 = 0$ and hence $\beta = 0$.

The only combination producing $\mathbf{0}$ is the trivial one, so $\mathbf{u}$ and $\mathbf{v}$ are independent. ∎

Remark (Where each hypothesis was used). Orthogonality killed the cross terms; being nonzero let us divide by $\|\mathbf{u}\|^2$. Drop the second hypothesis and the claim is false: $\mathbf{0}$ is orthogonal to every vector, since $\langle \mathbf{0}, \mathbf{v}\rangle = 0$ always, so $\{\mathbf{0}, \mathbf{v}\}$ is "mutually orthogonal" — and it is dependent, because $1 \cdot \mathbf{0} + 0 \cdot \mathbf{v} = \mathbf{0}$ uses a nonzero scalar. The word *nonzero* in the claim is not decoration.

21.3. The lemma: orthogonal vectors span the plane

Lemma. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ be nonzero and mutually orthogonal. Then

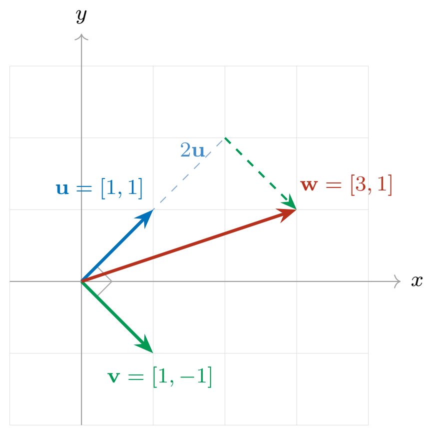
$$\operatorname{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2,$$

and $\{\mathbf{u}, \mathbf{v}\}$ is a basis for $\mathbb{R}^2$.

Proof. By §21.2 the pair is linearly independent. By §18.5, an independent pair in $\mathbb{R}^2$ spans $\mathbb{R}^2$; and by the uniqueness remark there, it is a basis. ∎

Two sentences, because both halves were already in hand. The content of the lemma is entirely in the link: a condition you can check with one multiplication-and-addition (§11.7) delivers a basis.

Example 21.1. $\mathbf{u} = [1, 1]$ and $\mathbf{v} = [1, -1]$ are nonzero, and $\langle \mathbf{u}, \mathbf{v}\rangle = 1(1) + 1(-1) = 0$. So without solving anything, they form a basis for $\mathbb{R}^2$.



Remark (Two is the most you can have). In $\mathbb{R}^2$ there is no such thing as three mutually orthogonal nonzero vectors. Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ were three of them. By the lemma $\operatorname{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2$, so $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$ for some scalars. Dot that with $\mathbf{w}$ itself:

$$\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle = \alpha\langle \mathbf{w}, \mathbf{u} \rangle + \beta\langle \mathbf{w}, \mathbf{v} \rangle = 0,$$

since $\mathbf{w}$ is orthogonal to both. So $\mathbf{w} = \mathbf{0}$, contradicting that it was nonzero. The dimension of the space caps how many mutually orthogonal directions it can hold — at 2 here, and at n in $\mathbb{R}^n$.

21.4. The same principle in $\mathbb{R}^3$

Nothing in §21.2 mentioned the number of coordinates. Run the identical argument on three vectors: if $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ are nonzero and mutually orthogonal, and

$$\alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w} = \mathbf{0},$$

then dotting with $\mathbf{u}$ annihilates the β and γ terms and leaves $\alpha\|\mathbf{u}\|^2 = 0$, so $\alpha = 0$. Dotting with $\mathbf{v}$ gives $\beta = 0$, and with $\mathbf{w}$ gives $\gamma = 0$. So the three are linearly independent, by exactly the same three lines.

Example 21.2. The standard basis $\{\vec{\imath}, \vec{\jmath}, \vec{k}\}$ of §20.1 is mutually orthogonal and each vector is nonzero, so it is independent — which we had assumed rather than checked. A less obvious triple:

$$\mathbf{u} = [\,1,\ 2,\ 2\,], \quad \mathbf{v} = [\,2,\ -2,\ 1\,], \quad \mathbf{w} = [\,2,\ 1,\ -2\,],$$

where $\langle \mathbf{u}, \mathbf{v} \rangle = 2 - 4 + 2 = 0$, $\langle \mathbf{u}, \mathbf{w} \rangle = 2 + 2 - 4 = 0$ and $\langle \mathbf{v}, \mathbf{w} \rangle = 4 - 2 - 2 = 0$. All three have norm 3. Mutually orthogonal and nonzero, hence independent.

Remark (What is still owed). Independence we have proved outright. To conclude that such a triple *spans* $\mathbb{R}^3$ we would need the $\mathbb{R}^3$ version of §18.5 — that any three independent vectors in $\mathbb{R}^3$ span it — and §18.5 was proved only for two vectors in $\mathbb{R}^2$, by an elimination that used the 2×2 quantity $D = u_1v_2 - u_2v_1$. The general statement, that n independent vectors in $\mathbb{R}^n$ form a basis, is true and we will prove it once we have the machinery for $n \times n$ systems. Until then, treat the spanning half in $\mathbb{R}^3$ as announced rather than established; the independence half is fully proved above and is the part the rest of this section uses.

21.5. The converse fails

Orthogonality gives independence. The reverse implication does not hold, and one example settles it.

Example 21.3. Let $\mathbf{u} = [\,1,\,0\,]$ and $\mathbf{v} = [\,1,\,1\,]$. They are independent: no scalar α satisfies $\alpha[1,0] = [1,1]$, since the second coordinate would need $\alpha \cdot 0 = 1$. But

$$\langle \mathbf{u}, \mathbf{v} \rangle = 1(1) + 0(1) = 1 \neq 0,$$

so they are not orthogonal — indeed $\cos(\mathbf{u}, \mathbf{v}) = 1/(1 \cdot \sqrt{2}) = 1/\sqrt{2}$, an angle of $45°$. The basis $\{[1,2],[2,3]\}$ of §18.1 is another: independent, but $\langle [1,2],[2,3] \rangle = 2 + 6 = 8 \neq 0$.

So the two conditions are not equivalent:

$$\text{mutually orthogonal (and nonzero)} \implies \text{linearly independent},$$

but not conversely. Orthogonality is the *strictly stronger* condition, and it is worth being precise about what the extra strength buys. Independence says no vector is redundant — none can be built from the others. Orthogonality says something sharper: no vector has *any* overlap with the others, not even a partial one. In the language of §12.4, the projection of one onto another is the zero vector. Independent-but-not-orthogonal vectors do lean on each other somewhat; they simply do not lean completely.

21.6. Why an orthogonal basis is worth having

Here is the practical payoff, and it is the reason orthogonality is pursued rather than merely noticed.

Against a general basis $\{\mathbf{u}, \mathbf{v}\}$, finding the coordinates of a vector $\mathbf{w}$ means solving a linear system — that was the work of §18.1. Against an *orthogonal* basis, no system is needed. Write $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$ and dot with $\mathbf{u}$; the β term dies, exactly as in §21.2, leaving

$$\langle \mathbf{u}, \mathbf{w} \rangle = \alpha\|\mathbf{u}\|^2, \qquad \text{so} \qquad \alpha = \frac{\langle \mathbf{u}, \mathbf{w} \rangle}{\|\mathbf{u}\|^2},$$

and symmetrically $\beta = \langle \mathbf{v}, \mathbf{w} \rangle / \|\mathbf{v}\|^2$. Each coordinate is computed *independently of the other*, by one dot product and one division.

Compare that expression with §12.4: it is precisely the coefficient appearing in $\mathrm{proj}_{\mathbf{u}}(\mathbf{w})$. So for an orthogonal basis,

$$\mathbf{w} = \mathrm{proj}_{\mathbf{u}}(\mathbf{w}) + \mathrm{proj}_{\mathbf{v}}(\mathbf{w}),$$

a vector is the sum of its own shadows on the basis directions. This is the general form of the remark in §16.4 about why coordinates were free to read off against $\{\vec{\imath}, \vec{\jmath}\}$: that basis was orthogonal, and orthogonality was doing the work all along — being aligned with the axes was incidental.

Example 21.4. Take the orthogonal basis $\mathbf{u} = [\,1,\,1\,]$, $\mathbf{v} = [\,1,\,-1\,]$ from §21.3 and the vector $\mathbf{w} = [\,3,\,1\,]$. Then $\|\mathbf{u}\|^2 = 2$ and $\|\mathbf{v}\|^2 = 2$, so

$$\alpha = \frac{\langle \mathbf{u}, \mathbf{w} \rangle}{2} = \frac{3+1}{2} = 2, \qquad \beta = \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{2} = \frac{3-1}{2} = 1.$$

Checking: $2[\,1,\,1\,] + 1[\,1,\,-1\,] = [\,2,\,2\,] + [\,1,\,-1\,] = [\,3,\,1\,]$. ✓ No system was solved — just two dot products, which is the head-to-tail walk drawn in §21.3.

Looking ahead. If an orthogonal basis is this convenient, the obvious question is whether one can always be manufactured: given any basis, can it be converted into an orthogonal one spanning the same space? It can, and the procedure is built out of exactly the piece we already have — subtract off the projection, as in §12.6, so that what remains is orthogonal to what came before. Repeating that step down a list of vectors is the Gram–Schmidt procedure.

22. Subspaces as lines and planes

With orthogonality settled, return to the subspaces of §19. The classification there was stated algebraically; it is worth saying what the answers look like.

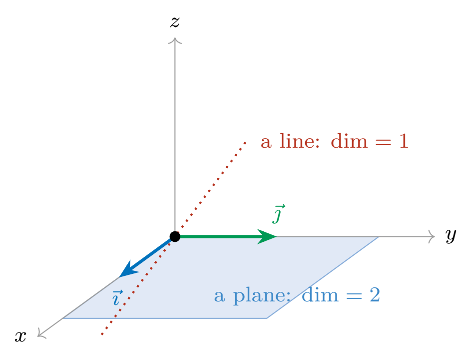
22.1. In $\mathbb{R}^2$: lines

Section 19.4 found exactly three kinds of subspace of $\mathbb{R}^2$: the single point $\{\mathbf{0}\}$, the whole plane, and everything in between — which turned out to be the lines through the origin. Two of those are degenerate: $\{\mathbf{0}\}$ has dimension 0 and $\mathbb{R}^2$ is not a proper subset. So the only interesting subspace of the plane is a **line through the origin**, of dimension 1, and it is $\operatorname{span}\{\mathbf{u}\}$ for any nonzero $\mathbf{u}$ lying on it. That was the line L of §18.6.

22.2. In $\mathbb{R}^3$: lines and planes

One dimension up, a new possibility appears. A subspace of $\mathbb{R}^3$ can be spanned by *two* independent vectors, and what they sweep out is a **plane through the origin**, of dimension 2.

We have already met one. The floor of §20.2 is $\operatorname{span}\{\vec{\imath}, \vec{\jmath}\}$: a plane through the origin, a subspace of $\mathbb{R}^3$, with basis $\{\vec{\imath}, \vec{\jmath}\}$ and dimension 2. Nothing about it was special to the axes — any two independent vectors in $\mathbb{R}^3$ span a plane through the origin, tilted however those two vectors happen to lie.



So the picture to carry is:

Dimension	**Subspace of $\mathbb{R}^2$**	**Subspace of $\mathbb{R}^3$**
0	the origin	the origin
1	a line through origin	a line through the origin
2	all of $\mathbb{R}^2$	a plane through the origin
3	—	all of $\mathbb{R}^3$

Read across: the same dimension means the same kind of object; what changes is whether that object exhausts the space it sits in. A plane is everything in $\mathbb{R}^2$ and a thin slice of $\mathbb{R}^3$.

A common confusion. "Line" and "plane" here always mean *through the origin.* The line $y = 2x+1$ and the plane $z = 1$ are not subspaces, for the reason given in §19.3: a subspace must contain $\mathbf{0}$, since scaling any of its elements by 0 produces it. A line or plane that misses the origin is a perfectly good geometric object, but it is not the span of anything and none of the machinery of this unit applies to it.

Looking ahead. The classification of subspaces of $\mathbb{R}^3$ — origin, lines, planes, everything — is proved exactly as §19.4 proved the $\mathbb{R}^2$ case: take a nonzero vector, its span is already inside; if that is not everything, take a vector outside it and enlarge; repeat. The only step not yet available is the one flagged in §21.4, that three independent vectors in $\mathbb{R}^3$ must span it. Once that is in place, the argument runs unchanged, and it will keep running in $\mathbb{R}^n$, where the subspaces are the origin, the lines, the planes, and so on up through dimension n.

23. The Pythagorean identity

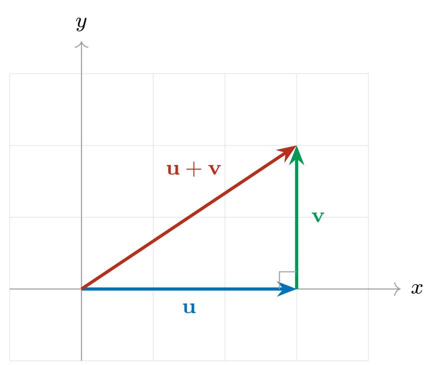
Orthogonality has one more consequence worth isolating, and it is the oldest theorem in the subject wearing new clothes.

23.1. The setup

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ be orthogonal, and consider their sum $\mathbf{u} + \mathbf{v}$. Read it head to tail, as in §7.1: walk along $\mathbf{u}$, then along $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$ is the single arrow from start to finish.

That walk traces a triangle — and this time the triangle has a right angle in it, because the two legs are orthogonal. So the three side lengths are $\|\mathbf{u}\|$, $\|\mathbf{v}\|$ and $\|\mathbf{u} + \mathbf{v}\|$, with the sum as the *hypotenuse*, and Pythagoras predicts

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{u} + \mathbf{v}\|^2.$$



Note that this is a genuine right triangle only because $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. For a general pair the head-to-tail triangle of §7.2 has no right angle and the identity is false — which is exactly why the triangle inequality there was an *inequality.*

23.2. The proof

Before proving it, recall the two collections of algebraic facts the computation will use. Vector addition and scalar multiplication obey the laws collected in §8.4 — commutativity, associativity, distributivity and the rest. The dot product obeys the laws collected in §11.8: it is symmetric, it distributes over addition, scalars pull out of it, and $\langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$. With those in hand the proof is four lines.

Start from the identity $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w}\rangle$ of §11.6, applied to $\mathbf{w} = \mathbf{u}+\mathbf{v}$, and expand by distributing twice:

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\|^2 &= \big\langle \mathbf{u}+\mathbf{v},\ \mathbf{u}+\mathbf{v}\big\rangle \\ &= \langle \mathbf{u},\mathbf{u}\rangle + \langle \mathbf{u},\mathbf{v}\rangle + \langle \mathbf{v},\mathbf{u}\rangle + \langle \mathbf{v},\mathbf{v}\rangle \\ &= \|\mathbf{u}\|^2 + 2\langle \mathbf{u},\mathbf{v}\rangle + \|\mathbf{v}\|^2,\end{aligned}$$

where the last step used symmetry, $\langle \mathbf{u},\mathbf{v}\rangle = \langle \mathbf{v},\mathbf{u}\rangle$, to combine the two middle terms. Now impose the hypothesis. Orthogonality says $\langle \mathbf{u},\mathbf{v}\rangle = 0$, so the middle term vanishes and

$$\boxed{\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2}$$

which is the identity. ■

Remark (Nothing here is new except the cancellation). The expansion $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u},\mathbf{v}\rangle + \|\mathbf{v}\|^2$ is not new — it is the computation already used at the end of §13 to derive the triangle inequality from Cauchy–Schwarz. It holds for *every* pair of vectors. The Pythagorean identity is that general expansion in the one case where the cross term $2\langle \mathbf{u},\mathbf{v}\rangle$ happens to be zero. So Pythagoras is not a separate theorem about right triangles; it is the special case of an algebraic identity, and the right angle is what kills the middle term.

Remark (It runs both ways). The expansion also gives the converse for free. If $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$, then comparing with the general expansion forces $2\langle \mathbf{u},\mathbf{v}\rangle = 0$, hence $\langle \mathbf{u},\mathbf{v}\rangle = 0$, hence $\mathbf{u}$ and $\mathbf{v}$ are orthogonal. So

$$\mathbf{u} \perp \mathbf{v} \qquad \Longleftrightarrow \qquad \|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2,$$

and the Pythagorean identity is not merely a consequence of orthogonality but a *characterisation* of it — one more entry alongside $\langle \mathbf{u},\mathbf{v}\rangle = 0$ and $\theta = \pi/2$.

Example 23.1. Take $\mathbf{u} = [\,3,\ 0\,]$ and $\mathbf{v} = [\,0,\ 4\,]$, which are orthogonal since $\langle \mathbf{u},\mathbf{v}\rangle = 0$. Then $\mathbf{u}+\mathbf{v} = [\,3,\ 4\,]$ and

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = 9 + 16 = 25 = \|[\,3,\ 4\,]\|^2.$$

The 3–4–5 triangle, recovered from the dot product.

For a pair not aligned with the axes, take $\mathbf{u} = [\,1,\ 1\,]$ and $\mathbf{v} = [\,1,\ -1\,]$ from §21.3. Then $\mathbf{u}+\mathbf{v} = [\,2,\ 0\,]$ and $2 + 2 = 4$. ✓

23.3. In $\mathbb{R}^3$ and beyond

The proof never mentioned coordinates, so it transfers untouched. For orthogonal $\mathbf{u},\mathbf{v} \in \mathbb{R}^3$ the same four lines give $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Example 23.2. From §21.4, $\mathbf{u} = [\,1,\ 2,\ 2\,]$ and $\mathbf{v} = [\,2,\ -2,\ 1\,]$ are orthogonal with $\|\mathbf{u}\| = \|\mathbf{v}\| = 3$. Their sum is $[\,3,\ 0,\ 3\,]$, and

$$\|\mathbf{u}+\mathbf{v}\|^2 = 9 + 0 + 9 = 18 = 9 + 9 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2. \quad ✓$$

It also extends past two vectors. If $\mathbf{u}_1, \ldots, \mathbf{u}_k$ are *mutually* orthogonal, expanding $\langle \mathbf{u}_1 + \cdots + \mathbf{u}_k,\ \mathbf{u}_1 + \cdots + \mathbf{u}_k\rangle$ produces one square term per vector and one cross term per pair — and every cross term is zero. So

$$\|\mathbf{u}_1 + \cdots + \mathbf{u}_k\|^2 = \|\mathbf{u}_1\|^2 + \cdots + \|\mathbf{u}_k\|^2.$$

Lengths do not add, but *squared* lengths do, provided the directions are mutually orthogonal.

23.4. An exact accounting of a projection

The identity earns its keep immediately, because §12.6 built an orthogonal pair for us. Recall that any $\mathbf{v}$ splits as

$$\mathbf{v} = \underbrace{\operatorname{proj}_{\mathbf{u}}(\mathbf{v})}_{\text{along } \mathbf{u}} + \underbrace{\big(\mathbf{v} - \operatorname{proj}_{\mathbf{u}}(\mathbf{v})\big)}_{\perp\, \mathbf{u}},$$

and that the second piece is orthogonal to $\mathbf{u}$ — hence orthogonal to the first piece, which is a multiple of $\mathbf{u}$. Two orthogonal vectors summing to $\mathbf{v}$: exactly the hypothesis of §23.2. So

$$\|\mathbf{v}\|^2 = \|\operatorname{proj}_{\mathbf{u}}(\mathbf{v})\|^2 + \|\mathbf{v} - \operatorname{proj}_{\mathbf{u}}(\mathbf{v})\|^2.$$

Read that as a budget. The squared length of $\mathbf{v}$ divides exactly into the part explained by $\mathbf{u}$ and the part left over, with nothing unaccounted for.

Remark (Cauchy–Schwarz, one more time). This sharpens §13. There we argued that a shadow is never longer than the object casting it, $|\operatorname{comp}_{\mathbf{u}}(\mathbf{v})| \le \|\mathbf{v}\|$, and turned it into Cauchy–Schwarz. The budget above proves the same inequality and says precisely what the slack is: since $\|\operatorname{proj}_{\mathbf{u}}(\mathbf{v})\| = |\operatorname{comp}_{\mathbf{u}}(\mathbf{v})|$, we get

$$\|\mathbf{v}\|^2 - \operatorname{comp}_{\mathbf{u}}(\mathbf{v})^2 = \|\mathbf{v} - \operatorname{proj}_{\mathbf{u}}(\mathbf{v})\|^2 \ \ge\ 0,$$

so the shadow falls short of the object by exactly the squared length of the orthogonal remainder — and equality holds precisely when that remainder is $\mathbf{0}$, which is when $\mathbf{v}$ already lies along $\mathbf{u}$. That is the equality case of Cauchy–Schwarz, recovered with the gap measured rather than merely bounded.

Looking ahead. Squared lengths adding over orthogonal pieces is the reason orthogonal decompositions are the workhorse of applied linear algebra. Split a vector into mutually orthogonal components and its squared length distributes across them without interference, so each component's contribution can be measured, ranked, and discarded independently of the others. That is the principle behind least squares, behind Parseval's identity in signal processing, and behind deciding how many components of a decomposition are worth keeping.